

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} S_{\text{IIB}} = & -T_{M_2} l \int d^2 \xi \sqrt{|\tilde{g}_{ij} - \frac{1}{k^2} (\partial_i \varrho + \tilde{B}_i^{(1)}) (\partial_j \varrho + \tilde{B}_j^{(1)})|} \\ & - \frac{T_{M_2} l}{2} \int d^2 \xi \epsilon^{ij} \left[\tilde{B}^{(2)}_{ij} + 2 \partial_i \varrho \left(\tilde{A}^{(1)}_i + m \pi \alpha' b_j \right) \right]. \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ccc} \{U_\alpha^1\} & \xleftarrow{\rho_{V, \{U_\alpha^1\}}} & \{U_\alpha^1|_V\} \\ \rho_{12} \uparrow & & \uparrow \rho_{V, 12} \\ \{U_\alpha^2\} & \xleftarrow{\rho_{V, \{U_\alpha^2\}}} & \{U_\alpha^2|_V\} \end{array}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} S_g = & \int d^4 x \, e \left\{ R(e, \omega) + 6g^2 + 2e^{-1} \epsilon^{\mu\nu\rho\sigma} \bar{\psi}_\mu \gamma_5 \gamma_\nu \left(\hat{\mathcal{D}}_\rho + ig A_\rho \sigma^2 \right) \psi_\sigma - \mathcal{F}^2 \right. \\ & \left. + \mathcal{J}_{(m)}{}^{\mu\nu} (\mathcal{J}_{(e)\mu\nu} + \mathcal{J}_{(m)\mu\nu}) \right\}, \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \|a\|_{\#} + \|x^{\natural} - a\|_{\#} &\geq \|x^{\natural}\|_{\#} \\ &\geq \|x^{\star}\|_{\#} \\ &= \|x^{\natural} + (x^{\star} - x^{\natural})\|_{\#} \\ &= \|a + (x^{\natural} - a) + z_1 + z_2\|_{\#} \\ &\geq \|a + z_1\|_{\#} - \|(x^{\natural} - a) + z_2\|_{\#} \\ &\geq \|a + z_1\|_{\#} - \|x^{\natural} - a\|_{\#} - \|z_2\|_{\#} \\ &= \|a\|_{\#} + \|z_1\|_{\#} - \|x^{\natural} - a\|_{\#} - \|z_2\|_{\#}. \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
A \circ_X B &= (AX)(X^\dagger B) \\
&= (AX^{2/3}X^{1/3})(X^{-1/3}X^{-2/3}B) \\
&= X^{1/3}[(X^{-1/3}AX^{2/3})(X^{-2/3}B)] \\
&= X^{1/3}[(X^{-1/3}AX^{1/3}X^{1/3})(X^{-1/3}X^{-1/3}B)] \\
&= X^{1/3}\{[(X^{-1/3}AX^{1/3})(X^{-1/3}BX^{1/3})]X^{-1/3}\} \\
&= X^{1/3}[(X^{-1/3}AX^{1/3})(X^{-1/3}BX^{1/3})]X^{-1/3}.
\end{aligned}$$